ANSWER KEY & STEP-BY-STEP SOLUTIONS

CLASS 8 – CBSE MATHEMATICS QUESTION PAPER

Maximum Marks: 30

SECTION A: SHORT ANSWER QUESTIONS (8 Marks)

Q1. Exponents & Powers

- **Formulas Used:**
  1. $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^{n}$
  2. $(a \cdot b)^{n} = a^{n} \cdot b^{n}$
  3. $\frac{a^{m}}{a^{n}} = a^{m-n}$
- **Step-by-Step Solution:**

Expression: $\left(\left(\frac{2}{3}\right)^{-2} \times \left(\frac{3}{4}\right)^{-3}\right) \div \left(\frac{8}{9}\right)^{-2}$

Step 1: Convert negative exponents to positive exponents by taking reciprocals:

$$\left(\left(\frac{3}{2}\right)^{2} \times \left(\frac{4}{3}\right)^{3}\right) \div \left(\frac{9}{8}\right)^{2}$$

Step 2: Express numbers as prime factors:

- $\frac{3}{2} = \frac{3}{2}$
- $\frac{4}{3} = \frac{2^{2}}{3}$
- $\frac{9}{8} = \frac{3^{2}}{2^{3}}$

Step 3: Substitute and expand powers:

$$\left(\frac{3^{2}}{2^{2}} \times \frac{(2^{2})^{3}}{3^{3}}\right) \div \frac{(3^{2})^{2}}{(2^{3})^{2}}$$

$$\left(\frac{3^{2}}{2^{2}} \times \frac{2^{6}}{3^{3}}\right) \div \frac{3^{4}}{2^{6}}$$

Step 4: Simplify the term inside the bracket:

$$\frac{2^{6-2}}{3^{3-2}} = \frac{2^{4}}{3^{1}} = \frac{16}{3}$$

Step 5: Perform division by multiplying with reciprocal:

$$\frac{2^{4}}{3^{1}} \times \frac{2^{6}}{3^{4}} = \frac{2^{4+6}}{3^{1+4}} = \frac{2^{10}}{3^{5}} = \left(\frac{2^{2}}{3}\right)^{5} = \left(\frac{\mathbf{4}}{\mathbf{3}}\right)^{\mathbf{5}} \quad \text{or} \quad \mathbf{\frac{1024}{243}}$$

- **Marking Scheme:**
  - **1 Mark:** Correct application of reciprocal rule for negative indices.

- **1 Mark:** Correct prime factorization, division simplification, and final exponential value.

Q2. Algebraic Identities

- **Formula Used:**

$$(a+b)^2 = a^2 + b^2 + 2ab$$

- **Step-by-Step Solution:**

Given: $x + \frac{1}{x} = 6$

Squaring both sides:

$$\left(x + \frac{1}{x}\right)^2 = (6)^2$$

$$x^2 + \left(\frac{1}{x}\right)^2 + 2 \cdot x \cdot \frac{1}{x} = 36$$

$$x^2 + \frac{1}{x^2} + 2 = 36$$

$$x^2 + \frac{1}{x^2} = 36 - 2 = \mathbf{34}$$

- **Marking Scheme:**
  - **1 Mark:** Writing the expansion identity $(x + 1/x)^2 = x^2 + 1/x^2 + 2$.
  - **1 Mark:** Substituting 6 and obtaining the final answer 34.

Q3. Direct & Inverse Variations

- **Formula Used:**

$$\text{Total Work} = W_1 \times D_1 \times H_1 = W_2 \times D_2 \times H_2 \quad \text{(Compound Proportion Formula)}$$

- **Step-by-Step Solution:**

Given:

- Case 1: $W_1 = 12$ workers, $D_1 = 20$ days, $H_1 = 8$ hours/day
- Case 2: $W_2 = 16$ workers, $D_2 = x$ days, $H_2 = 6$ hours/day

Equating work units:

$$12 \times 20 \times 8 = 16 \times x \times 6$$

$$1920 = 96x$$

$$x = \frac{1920}{96} = \mathbf{20} \text{ days}$$

- **Marking Scheme:**
  - **1 Mark:** Establishing inverse proportion relation / product formula $W_1 D_1 H_1 = W_2 D_2 H_2$.
  - **1 Mark:** Correct algebraic computation obtaining 20 days.

Q4. Factorisation

- **Formula Used:**

$$a^2 - b^2 = (a - b)(a + b)$$

- **Step-by-Step Solution:**

Expression: $16x^4 - 81y^4$

Step 1: Rewrite as difference of squares:

$$(4x^2)^2 - (9y^2)^2$$

Step 2: Apply identity $a^2 - b^2$:

$$= (4x^2 - 9y^2)(4x^2 + 9y^2)$$

Step 3: Further factorise the first term $(4x^2 - 9y^2)$:

$$4x^2 - 9y^2 = (2x)^2 - (3y)^2 = (2x - 3y)(2x + 3y)$$

Step 4: Combine all factors:

$$= \mathbf{(2x - 3y)(2x + 3y)\left(4x^2 + 9y^2\right)}$$

- **Marking Scheme:**
  - **1 Mark:** First level factorisation into $(4x^2 - 9y^2)(4x^2 + 9y^2)$.
  - **1 Mark:** Complete breakdown of $(4x^2 - 9y^2)$ into $(2x - 3y)(2x + 3y)$.

SECTION B: MEDIUM ANSWER QUESTIONS (12 Marks)

Q5. Mensuration (Trapezium & Volume)

- **Formulas Used:**
  1. $A_{\text{trapezium}} = \frac{1}{2}(a + b)h$
  2. $V_{\text{prism}} =$ Cross-sectional Area $\times$ Length
- **Step-by-Step Solution:**

(a) **Cross-sectional Area:**

  - $a = 12$ m, $b = 8$ m, $h = 4$ m

$$A = \frac{1}{2} \times (12 + 8) \times 4 = \frac{1}{2} \times 20 \times 4 = \mathbf{40}\ \text{m}^{\mathbf{2}}$$

(b) **Volume of Earth Dug Out:**

  - Length $(L) = 250$ m

$$V = A \times L = 40\ \text{m}^2 \times 250\ \text{m} = \mathbf{10,000}\ \text{m}^{\mathbf{3}}$$

- **Marking Scheme:**
  - **1.5 Marks:** Formula substitution and calculation for area ($40\ \text{m}^2$).
  - **1.5 Marks:** Multiplying area by length to get volume ($10{,}000\ \text{m}^3$).

Q6. Comparing Quantities (Successive Depreciation)

- **Formula Used:**

$$A = P\left(1 - \frac{R_1}{100}\right)\left(1 - \frac{R_2}{100}\right)$$

- **Step-by-Step Solution:**

Given:

  - Principal / Initial Value (P) = Rs. $1{,}25{,}000$
  - Rate $R_1 = 8\%$, Rate $R_2 = 10\%$

(a) **Depreciated Value (A):**

$$A = 125000\left(1 - \frac{8}{100}\right)\left(1 - \frac{10}{100}\right)$$

$$A = 125000 \times \left(\frac{92}{100}\right) \times \left(\frac{90}{100}\right)$$

$$A = 125000 \times 0.92 \times 0.9 = 125000 \times 0.828 = \text{Rs. } \mathbf{1,03,500}$$

(b) **Total Depreciation Amount:**

$$\text{Depreciation} = P - A = 125000 - 103500 = \text{Rs. } \mathbf{21,500}$$

- **Marking Scheme:**
  - **2 Marks:** Substitution in successive depreciation formula and calculating $A =$ Rs. $1{,}03{,}500$.
  - **1 Mark:** Subtracting depreciated value from initial price to get Rs. $21{,}500$.

Q7. Algebraic Expansion & Simplification

- **Formulas Used:**
  1. $(a+b)^2 - (a-b)^2 = 4ab$
  2. $(a-b)(a+b) = a^2 - b^2$
- **Step-by-Step Solution:**

$$\text{Expression: } [(3x+4y)^2 - (3x-4y)^2] + (3x-4y)(3x+4y)$$

Part 1: Expand $(3x+4y)^2 - (3x-4y)^2$:

$$= (9x^2 + 24xy + 16y^2) - (9x^2 - 24xy + 16y^2)$$

$$= 9x^2 + 24xy + 16y^2 - 9x^2 + 24xy - 16y^2 = 48xy$$

Part 2: Expand $(3x-4y)(3x+4y)$:

$$= (3x)^2 - (4y)^2 = 9x^2 - 16y^2$$

Part 3: Combine both parts:

$$= 48xy + 9x^2 - 16y^2 = \mathbf{9x^2 + 48xy - 16y^2}$$

- **Marking Scheme:**
  - **1.5 Marks:** Correct simplification of $[(3x+4y)^2 - (3x-4y)^2]$ to obtain $48xy$.
  - **1.5 Marks:** Expanding $(3x-4y)(3x+4y)$ to $9x^2 - 16y^2$ and writing the final simplified expression.

Q8. Solid Geometry (Hollow Cylinder)

- **Formulas Used:**
  1. $V = \pi(R^2 - r^2)h$
  2. $TSA = 2\pi Rh + 2\pi rh + 2\pi(R^2 - r^2)$
- **Step-by-Step Solution:**

Given: Outer radius $R = 8$ cm, Inner radius $r = 6$ cm, Height $h = 14$ cm, $\pi = \frac{22}{7}$

(a) **Volume of Metal:**

$$R^2 - r^2 = 8^2 - 6^2 = 64 - 36 = 28 \text{ cm}^2$$

$$V = \frac{22}{7} \times 28 \times 14 = 22 \times 4 \times 14 = \mathbf{1232} \text{ cm}^{\mathbf{3}}$$

(b) **Total Surface Area (TSA):**

- Outer Curved Area = $2\pi Rh = 2 \times \frac{22}{7} \times 8 \times 14 = 704 \text{ cm}^2$
- Inner Curved Area = $2\pi rh = 2 \times \frac{22}{7} \times 6 \times 14 = 528 \text{ cm}^2$
- Area of 2 Base Rings = $2\pi(R^2 - r^2) = 2 \times \frac{22}{7} \times 28 = 176 \text{ cm}^2$

$$TSA = 704 + 528 + 176 = \mathbf{1408} \text{ cm}^{\mathbf{2}}$$

- **Marking Scheme:**
  - **2 Marks:** Volume computation formula and exact value (1232 cm^3).
  - **1 Mark:** Total Surface Area calculation (1408 cm^2).

SECTION C: LONG ANSWER & MULTI-FORMULA PROBLEMS (10 Marks)

Q9. Multi-Shape Mensuration & Cost Computation

- **Formulas Used:**
  1. $V_{\text{total}} = \pi r^2 h + \frac{2}{3}\pi r^3$
  2. $A_{\text{internal}} = 2\pi rh + 2\pi r^2$
  3. Cost = Area × Rate
- **Step-by-Step Solution:**

Given: Radius $r = 7$ m, Cylinder height $h = 15$ m, $\pi = \frac{22}{7}$

(a) **Total Volume:**

$$V_{\text{cylinder}} = \pi r^2 h = \frac{22}{7} \times 7^2 \times 15 = 22 \times 7 \times 15 = 2310 \text{ m}^3$$

$$V_{\text{hemisphere}} = \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 7^3 = \frac{2}{3} \times 22 \times 49 = \frac{2156}{3} \approx 718.67 \text{ m}^3$$

$$V_{\text{total}} = 2310 + 718.67 = \mathbf{3028.67} \text{ m}^{\mathbf{3}} \quad \left(\text{or } \frac{9086}{3} \text{ m}^3\right)$$

(b) **Total Internal Surface Area:**

$$S_{\text{cylinder}} = 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 15 = 660 \text{ m}^2$$

$$S_{\text{hemisphere}} = 2\pi r^2 = 2 \times \frac{22}{7} \times 7^2 = 308 \text{ m}^2$$

$$S_{\text{total}} = 660 + 308 = \mathbf{968} \text{ m}^2$$

(c) **Cost of Painting:**

$$\text{Cost} = 968 \text{ m}^2 \times \text{Rs. } 45/\text{m}^2 = \text{Rs. } \mathbf{43,560}$$

- **Marking Scheme:**
  - **2.5 Marks:** Individual volume computations and total volume calculation (3028.67 m^3).
  - **1.5 Marks:** Summing curved surface areas to get 968 m^2.
  - **1 Mark:** Multiplying area by rate (Rs. $43{,}560$).

Q10. Advanced Financial Mathematics (Compound Interest Differentials)

- **Formulas Used:**
  1. **Compounded Annually:** $A = P\left(1 + \frac{R}{100}\right)^n$

(For fractional time $1\frac{1}{2}$ years: $A = P\left(1 + \frac{R}{100}\right)^1 \times \left(1 + \frac{R/2}{100}\right)^1$)

  2. **Compounded Half-Yearly:** $A = P\left(1 + \frac{R/2}{100}\right)^{2n}$

- **Step-by-Step Solution:**

Given: $P =$ Rs. $80{,}000$, $R = 10\%$ p.a., $t = 1.5$ years

(a) **Compounded Annually:**

  - For 1 full year:

$$A_1 = 80000 \times \left(1 + \frac{10}{100}\right) \times \left(1 + \frac{5}{100}\right)$$

$$A_1 = 80000 \times \frac{11}{10} \times \frac{21}{20} = 80000 \times \frac{231}{200} = 400 \times 231 = \text{Rs. } \mathbf{92,400}$$

  - Interest $CI_1 = A_1 - P = 92400 - 80000 = \text{Rs. } \mathbf{12,400}$

(b) **Compounded Half-Yearly:**

  - Number of half-years ($2n$) = $1.5 \times 2 = 3$ conversion periods
  - Rate per half-year (R') = $\frac{10\%}{2} = 5\%$

$$A_2 = 80000 \times \left(1 + \frac{5}{100}\right)^3 = 80000 \times \left(\frac{21}{20}\right)^3$$

$$A_2 = 80000 \times \frac{9261}{8000} = 10 \times 9261 = \text{Rs. } \mathbf{92,610}$$

  - Interest $CI_2 = A_2 - P = 92610 - 80000 = \text{Rs. } \mathbf{12,610}$

(c) **Difference in Interest Earned:**

$$\text{Difference} = CI_2 - CI_1 = 12610 - 12400 = \text{Rs. } \mathbf{210}$$

- **Marking Scheme:**
  - **2 Marks:** Annual compounding calculations ($A_1 = \text{Rs. } 92{,}400$, $CI_1 = \text{Rs. } 12{,}400$).
  - **2 Marks:** Half-yearly compounding calculations ($A_2 = \text{Rs. } 92{,}610$, $CI_2 = \text{Rs. } 12{,}610$).
  - **1 Mark:** Correct difference calculation (Rs. 210).